Dialect-aware Beam Search

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Abstract

Off-target translation remains a major challenge for low-resource dialectal machine translation, yet existing approaches primarily address it during model training. In this work, we present a qualitative analysis of off-target dialect translation during beam search decoding using the Dari and Farsi dialects of Persian as a case study. By analyzing the beam candidates generated by two state-of-the-art multilingual decoder-only LLMs, we find that the correct target-dialect token often already exists within the beam but is assigned a lower rank than the selected standard-dialect lexical variant. Our findings provide insights into how off-target dialect translations emerge during decoding and motivate future decoding-time methods for improving low-resource dialectal machine translation. Related codes are available at https://github.com/waisyousofi/DialB.

1 Introduction

Machine translation for dialects of a language has received increasing attention from the Natural Language Processing (NLP) community due to its potential to improve linguistic inclusivity and extend machine translation technologies beyond standardized language varieties (Joshi et al., 2025). In this work, we use the term *dialect* as an umbrella term encompassing regional, national, and sociolinguistic varieties of a language. These varieties are characterized by systematic lexical, syntactic, morphological, and orthographic differences while remaining mutually intelligible to varying degrees among the speakers of the the corresponding dialects of a language.

Such linguistic variations present unique challenges for machine translation models requiring them to generate translations that not only preserve the source meaning but must also use the lexical and syntactic forms of the intended target dialect. State-of-the-art machine translation models are predominantly based on the Transformer (Vaswani et al., 2017) architecture, with multilingual Neural Machine Translation (MNMT) models becoming the standard approach for translating between multiple languages and dialects. Their multilingual nature enables transfer learning across related language varieties, partially alleviating the scarcity of parallel data for MT between low-resource languages and dialects. However, recent studies have shown that, despite these advances, MNMT models generate a substantial number of off-target translations, particularly in low-resource and zero-shot translation directions (Zhang et al., 2020; Yang et al., 2021). These are the problems when the generated output contains tokens from the source dialect, the corresponding standard language, or even from other languages instead of the intended target dialect.

To the best of our knowledge, with the exception LiBS (Yang et al., 2024), which addresses off-target language translation at the decoding stage of test time, all existing approaches are motivated to mitigate off-target translation during the training stage. Similar to LiBS, our work focuses on the decoding stage of off-the-shelf pretrained multilingual decoder-only Large Language Models (LLMs). Specifically, we undertake a case study of machine translation among the Dari (Afghan Persian) and Farsi (Iranian Persian) dialects of the Persian language, a dialect pair that has not been studied before. Using beam search which is the most widely employed decoding strategy in LLMs, we are interested to qualitatively analyze off-target dialect translations and understand whether the correct target-dialect tokens exist within the beams despite not being ranked as the most likely candidates. Based on this analysis, we want to identify possible ways to increase the likelihood of selecting the correct target-dialect tokens during decoding, thereby providing insights and future research directions for reducing off-target dialect translation.

Our study is conducted using two state-of-the-art pretrained multilingual decoder-only LLMs, Gemma-4-31B-it and TranslateGemma-27B-it, to investigate whether the observed decoding behaviour is consistent across different models. We

conduct our experiments on two publicly available benchmark datasets, FLORES-200 and CoDET (NLLB Team et al., 2024; Alam et al., 2024). Due to quality issues observed in the Persian dialect references of FLORES-200, our qualitative analysis is performed primarily on CoDET. The qualitative analysis is carried out manually by the first author (*), a native Persian speaker fluent in both Dari and Farsi.

2 Related Work

To the best of our knowledge, LiBS (Yang et al., 2024) is the first work to investigate off-target translation at the decoding stage. The authors analyze how off-target language translations emerge during beam search decoding and propose Language-informed Beam Search (LiBS), an inference-time decoding algorithm that incorporates an external language identification model to reduce off-target language generation.

In our work, we draw inspiration from LiBS to investigate off-target translation in the low-resource setting of DMT. Specifically, we aim to understand how off-target dialect translations emerge during decoding and identify potential opportunities for improving decoding in low-resource dialectal MT.

3 Experiment

3.1 Dataset

We conduct our study on the CoDET benchmark (Alam et al., 2024), a publicly available benchmark for dialect evaluation in machine translation. Among the many dialects included in the benchmark, we consider only the Persian dialects—Dari and Farsi. The dataset consists of 3,071 manually translated English sentence pairs created by community volunteers and belongs to the medical (COVID-19) domain. The average sentence length is approximately 27 words for both the Dari and Farsi translations.

3.2 Baseline Models

We conduct our analysis using two state-of-the-art multilingual decoder-only LLMs: Gemma-4-31B-it and TranslateGemma-27B-it. Gemma-4-31B-it serves as a strong general-purpose multilingual LLM, while TranslateGemma-27B-it is specifically designed for multilingual translation. Both models are used in their off-the-shelf pretrained form without any task-specific fine-tuning or parameter updates. Unless otherwise stated, beam search is used as the decoding strategy for all experiments.

Table 1: Baseline translation performance on the CoDET benchmark. Beam search consistently outperforms greedy decoding across both translation directions.

Direction	Decoding	BLEU	ChrF++
Farsi → Dari	Greedy	10.98	36.52
	Beam	**11.13**	**36.85**
Dari → Farsi	Greedy	19.36	46.95
	Beam	**19.54**	**47.18**

3.3 Baseline Results

We first compare greedy decoding and beam search on the CoDET benchmark using BLEU and ChrF++. As shown in Table 1, beam search consistently outperforms greedy decoding in both translation directions, although the improvements are modest. Since beam search is also the most widely adopted decoding strategy for autoregressive LLMs, we use beam search for all subsequent qualitative analyses.

3.4 Analysis and Results

Since beam search consistently outperforms greedy decoding on both translation directions (Table 1), our qualitative analysis focuses on translations generated using beam search. For each off-target translation, we manually examine the generated output together with the beam candidates at every decoding step to determine whether the correct target-dialect token was generated by the model but assigned a lower rank than the selected token.

Our analysis of the Farsi→Dari translation direction reveals a recurring pattern. In many cases, the correct Dari token already exists within the beam, while the corresponding standard Farsi lexical variant is selected instead. This observation suggests that a considerable proportion of off-target dialect translations arise from token ranking during decoding rather than from the inability of the pretrained model to generate the correct dialectal token. Several representative examples are presented in Appendix B.

4 Conclusions

In this work, we presented a qualitative study of off-target dialect translation during beam search decoding for low-resource dialectal machine translation. Using the Dari and Farsi dialects of Persian as

a case study, we analyzed the decoding behaviour of two state-of-the-art multilingual decoder-only LLMs, Gemma-4-31B-it and TranslateGemma-27B-it. Our analysis shows that, in many cases, the correct target-dialect token already exists within the beam but is assigned a lower rank than the selected standard-dialect lexical variant. This finding suggests that a considerable proportion of off-target dialect translations arise from token ranking during decoding rather than from the model's inability to generate the correct dialectal token. We hope that the insights presented in this work will facilitate future research on decoding-time methods for improving translation quality in low-resource dialectal machine translation.

5 Future Work

Our qualitative analysis suggests that a considerable proportion of off-target dialect translations arise from token ranking during decoding rather than from the inability of the model to generate the correct target-dialect token. One possible reason is the limited quality of the tokenizer for low-resource dialects. We observed that many dialect-specific words are segmented into individual characters or linguistically meaningless subword units instead of meaningful morphemes. Such fragmented representations may lead to lower token probabilities during decoding, preventing the correct dialectal tokens from being ranked among the most likely candidates. An immediate direction for future work is therefore to develop improved tokenizers for low-resource Persian dialects by leveraging large monolingual dialectal corpora.

Inspired by LiBS, which incorporates an external language identification model during decoding, another promising direction is to investigate the incorporation of dialect-aware perplexity scores during beam search. We plan to fine-tune a dialect-specific language model on large monolingual dialectal corpora and use its perplexity to guide beam ranking toward the intended target dialect.

Finally, although this work focuses on translation between the Dari and Farsi dialects of Persian, we plan to extend our analysis to Tajik in future work, enabling a broader investigation of decoding behaviour across the three major Persian dialects.

References

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A Prompt and Inference Configuration

For all experiments, we use the following instruction prompt. The placeholders `{parent_lang}`, `{sd}`, `{td}`, and `{source_sentence}` are replaced with the corresponding language, source dialect, target dialect, and input sentence for each translation instance. Table 2 summarizes the inference configuration used throughout the experiments.

```
### Instruction:
You are an expert in {parent_lang} dialect translation.
Translate the following sentence from {sd} to {td},
preserving meaning while adapting vocabulary,
morphology, syntax, and dialect-specific usage to
natural {td}.

Input:
{source_sentence}

Return only the translated sentence without any
explanation.

### Response:
```

Table 2: Inference configuration used in all experiments.

Parameter	Value
Models	Gemma-4-31B-it, TranslateGemma-27B-it
Decoding Strategy	Beam Search
Number of Beams	4
Top-k Candidates	10
Maximum New Tokens	128
Length Penalty	1.0

B Qualitative Error Analysis Examples

This appendix presents representative examples of off-target dialect translation produced by Gemma-4-31B-it. All Persian sentences are romanized for readability. For each example, we show the source sentence, English translation, model output, and the dialect-specific lexical substitutions identified during manual analysis.

B.1 Gemma-4-31B-it

Example 1: Standard Farsi lexical substitutions

Sentence Index: 2025

Farsi Dialectal Word(s):

havapeyma; bimar

Input Sentence (Romanized Farsi):

khadame-ye havapeyma bayad moghe'-e moraghebat az mosafer-e bimar ya moghe'-e tamas ba maye'at-e badan ya sotuh-e aludeh, az dastkesh-ha-ye pezeshki-ye yekbar-masraf estefadeh konad va dar surati ke fard-e bimar tab, sorfeh-ye modavem ya moshkel-e tanaffos dashteh bashad, bayad az tajhizat-e moharezat-e shakhsi-ye bishtari estefadeh konand.

Input in English:

Cabin crew should wear disposable medical gloves when tending to a sick traveler or touching body fluids or potentially contaminated surfaces, and possibly additional personal protective equipment if the sick traveler has fever, persistent cough, or difficulty breathing.

Model Output (Romanized):

sarbazan-e havapeyma bayad hengam-e moraghebat az mosafer-e bimar ya hengam-e tamas ba maye'at-e badan ya sotuh-e aludeh, az dastkesh-ha-ye tebbi-ye yekbar-masraf estefadeh konand va dar surati ke fard-e bimar tab, sorfeh-ye modavem ya moshkel-e tanaffosi dashteh bashad, bayad az tajhizat-e hefazati-ye shakhsi-ye bishtari estefadeh konand.

Erroneous Token(s):

sarbazan-e havapeyma; bimar

Correct Token(s):

khadame-ye kabin; mariz

Correct Token Exists in Beam: Yes
Correct Token Exists in Reference: Yes

Example 2: Standard Farsi word selected instead of Dari equivalent

Sentence Index: 2024

Farsi Dialectal Word:

bimar

Input Sentence (Romanized Farsi):

agar fardi dar dakhel-e havapeyma bimar shavad, ravesh-ha-ye monaseb baraye mohafezat az karkonan va sa'er mosaferan mitavanad shamel-e joda kardan-e fard-e bimar az digaran be fasele-ye 6 foot, takhsis-e tanha yeki az khadameh baraye khadamat-dehi be bimar va era'e-ye mask-e surat be bimar, ya darkhast az bimar baraye pushandan-e dahan va bini-ash hengam-e sorfeh ya atseh, bashad.

Input in English:

If a person becomes sick on an airplane, proper controls to protect workers and other passengers include separating the sick person from others by a distance of 6 feet, designating one crew member to serve the sick person, and offering a face mask to the sick person or asking the sick person to cover their mouth and nose with tissues when coughing or sneezing.

Model Output (Romanized):

agar kasi dar dakhel-e tayyara bimar shavad, ravesh-ha-ye monaseb baraye mohafezat az karkonan va sa'er mosaferan mitavanad shamel-e joda kardan-e shakhs-e bimar az digaran be fasele-ye 6 foot, ta'yin-e tanha yeki az khadameh baraye residagi be bimar va dadan-e mask-e surat be u, ya darkhast az bimar baraye pushandan-e dahan va bini-ash hengam-e sorfeh ya atseh bashad.

Erroneous Token:

bimar

Correct Token:

mariz

Correct Token Exists in Beam: Yes
Correct Token Exists in Reference: Yes

Example 3: Standard Farsi word selected instead of Dari equivalent

Sentence Index: 1141

Farsi Dialectal Word:

ta'tilat

Input Sentence (Romanized Farsi):

Domain.com yeki az sayt-ha-ye fa'al dar zamine-ye amlak gozaresh dad ke hengami ke be dalil-e ta'tili-ye kharidaran dar ta'tilat-e tulani-ye akhar-e hafte-ye ruz-e kargar, mozayede-ha-ye kamtari bargozar mishod, barkhi forushandegan-e amlak be avalin pishnahad-dahandegan-e mozayede-ha dar Melbourne, dastmal-e toalet-e rayegan ehda mikardand.

Input in English:

Domain.com, a real estate site, reported some property sellers offering free toilet paper to the first bidder on auctions in Melbourne, when fewer auctions were held because of buyers having time off on the long weekend of Labour Day.

Model Output (Romanized):

sayt-e Domain.com ke yeki az website-ha-ye fa'al dar zamine-ye amlak ast, gozaresh dad ke dar jaryan-e ta'tilat-e tulani-ye akhar-e hafte-ye ruz-e kargar, be dalil-e ta'tili va kahesh-e tedad-e mozayede-ha, barkhi az forushandegan-e amlak dar Melbourne be avalin pishnahad-dahandegan-e mozayede, dastmal-e toalet-e rayegan midadand.

Erroneous Token:

ta'tilat

Correct Token:

rukhsati

Correct Token Exists in Beam: Yes
Correct Token Exists in Reference: Yes

B.2 TranslateGemma-27B-it

Example 1: Standard Farsi lexical substitutions

Sentence Index: 2025

Farsi Dialectal Word(s):

havapeyma; bimar

Input Sentence (Romanized Farsi):

khadame-ye havapeyma bayad moghe'-e moraghebat az mosafer-e bimar ya moghe'-e tamas ba maye'at-e badan ya sotuh-e aludeh, az dastkesh-ha-ye pezeshki-ye yekbar-masraf estefadeh konad va dar surati ke fard-e bimar tab, sorfeh-ye modavem ya moshkel-e tanaffos dashteh bashad, bayad az tajhizat-e moharezat-e shakhsi-ye bishtari estefadeh konand.

Input in English:

Cabin crew should wear disposable medical gloves when tending to a sick traveler or touching body fluids or potentially contaminated surfaces, and possibly additional personal protective equipment if the sick traveler has fever, persistent cough, or difficulty breathing.

Model Output (Romanized):

khadame-ye havapeyma hengam-e moraghebat az mosaferan-e bimar ya hengam-e tamas ba maye'at-e badan ya sotuh-e aludeh, bayad az dastkesh-ha-ye pezeshki-ye yekbar-masraf estefadeh konand. Hamchenin, dar surati ke mosafer-e bimar tab, sorfeh-ye modavem ya moshkel-e tanaffosi dashteh bashad, bayad az tajhizat-e moharezat-e shakhsi-ye bishtari estefadeh konand.

Erroneous Token(s):

havapeyma; bimar

Correct Token(s):

khadame-ye parvaz / kabin; mariz

Correct Token Exists in Beam: Yes
Correct Token Exists in Reference: Yes

Example 2: Standard Farsi lexical substitution

Sentence Index: 59
Farsi Dialectal Word:

bastegan

Input Sentence (Romanized Farsi):

aya bastegan-e shoma ham hamin alayem ra darand?

Input in English:

Do your relatives have the same symptoms?

Model Output (Romanized):

aya a'za-ye khanevade-ye shoma ham alayem-e moshabehi darand?

Erroneous Token:

a'za-ye khanevade

Correct Token(s):

nazdikan / aqarib

Correct Token Exists in Beam: Yes
Correct Token Exists in Reference: Yes